

Practical week - 6

routes/catalog.py :

```
from fastapi import APIRouter, HTTPException
```

```
router = APIRouter()
```

```
books = [
```

```
{
```

```
    "id": 101,
```

```
    "title": "Python Programming",
```

```
    "author": "John Smith",
```

```
    "price": 450,
```

```
    "isbn": "9781234567890",
```

```
    "status": "Available"
```

```
},
```

```
{
```

```
    "id": 102,
```

```
    "title": "Database Management",
```

```
    "author": "Robert Brown",
```

```
    "price": 550,
```

```
    "isbn": "9789876543210",
```

```
    "status": "Available"
```

```
}
```

```
]
```

```
@router.get("/books") def
```

```
get_books():
```

```
    return books
```

```
@router.get("/books/{book_id}")
```

```
def get_book(book_id: int):
```

```
    for book in books:
```

```
        if book["id"] == book_id:
```

```
            return book raise
```

HTTPException(

```
        status_code=404,  
        detail="Book not found"  
    )
```

routes/admin.py :

```
from fastapi import APIRouter, HTTPException  
from pydantic import BaseModel, Field  
from routes.catalog import books  
router = APIRouter()
```

```
class Book(BaseModel):
```

```
    title: str  
    author: str  
    price: float = Field(ge=0)  
    isbn: str  
    status: str
```

```
@router.post("/admin/books") def
```

```
add_book(book: Book):  
    new_id = max([b["id"] for b in books], default=100) + 1  
    new_book = {  
        "id": new_id,  
        book.model_dump()  
    }  
    books.append(new_book)  
    return {  
        "message": "Book added successfully",  
        "book": new_book  
    }
```

```
@router.put("/admin/books/{book_id}")
```

```
def update_book(book_id: int, book: Book):  
    for index, existing_book in enumerate(books):  
        if existing_book["id"] == book_id:
```

```
updated_book = {
```

```

        "id": book_id,
        book.model_dump()
    }
    books[index] = updated_book
    return {
        "message": "Book updated successfully",
        "book": updated_book
    }

    raise HTTPException(
        status_code=404,
        detail="Book not found"
    )

@router.delete("/admin/books/{book_id}")
def delete_book(book_id: int):
    for index, book in enumerate(books):
        if book["id"] == book_id:
            deleted_book = books.pop(index)
            return {
                "message": "Book deleted successfully",
                "book": deleted_book
            }

    raise HTTPException(
        status_code=404,
        detail="Book not found"
    )

```

main.py :

```

from fastapi import FastAPI

from routes.catalog import router as catalog_router
from routes.admin import router as admin_router

app = FastAPI(
    title="BookFlow Library API"
)

app.include_router(catalog_router)
app.include_router(admin_router)

```

GET default all the data:

default

GET /books Get Books

Parameters

No parameters

Execute

Responses

Curl

Request URL

Server response

Code

Details

200

Response body

Response headers

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json'
```

```
http://127.0.0.1:8000/books
```

```
200
Response body
{
  "id": 101,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 499,
  "isbn": "9781234567890",
  "status": "Available"
},
{
  "id": 102,
  "title": "Database Management",
  "author": "Robert Brown",
  "price": 350,
  "isbn": "9780987654321",
  "status": "Available"
}
Response headers
content-length: 280
content-type: application/json
date: Tue, 15 Sep 2020 06:05:14 GMT
server: uvicorn
```

GET:

GET /books/{book_id} Get Book

Parameters

Name

Description

book_id * required

integer

(path)

101

Execute

Response:

Responses

Curl

Request URL

Server response

Code

Details

200

Response body

Response headers

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books/101' \
  -H 'accept: application/json'
```

```
http://127.0.0.1:8000/books/101
```

```
200
Response body
{
  "id": 101,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 499,
  "isbn": "9781234567890",
  "status": "Available"
}
Response headers
content-length: 117
content-type: application/json
date: Tue, 15 Sep 2020 06:22:00 GMT
server: uvicorn
```

POST:

POST /admin/books Add Book

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "title": "Java Programming",
  "author": "Chaitanya",
  "price": 680,
  "isbn": "9781234567891",
  "status": "Available"
}
```

Execute

Response:

Responses

Curl

```
curl -X 'POST' \
  http://127.0.0.1:8000/admin/books' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Java Programming",
    "author": "Chaitanya",
    "price": 600,
    "isbn": "9781234567891",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code

Details

200

Response body

```
{
  "message": "Book added successfully",
  "book": {
    "id": 10,
    "title": "Java Programing",
    "author": "Chaitanya",
    "price": 600,
    "isbn": "9781234567891",
    "status": "Available"
  }
}
```

Response headers

```
content-length: 160
content-type: application/json
date: Tue, 15 Sep 2020 06:34:18 GMT
server: uvicorn
```

PUT:

PUT

/admin/books/{book_id} Update Book

Parameters

Name	Description
book_id * required	
integer	101
(path)	

Request body required

Edit Value | Schema

```
{
  "title": "JavaScript Essentials",
  "author": "John Smith",
  "price": 450,
  "isbn": "9781234567890",
  "status": "Available"
}
```

Response:

Responses

Curl

```
curl -X 'PUT' \
  http://127.0.0.1:8000/admin/books/101' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "JavaScript Essentials",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books/101

Server response

Code

Details

200

Response body

```
{
  "message": "Book updated successfully",
  "book": {
    "id": 101,
    "title": "JavaScript Essentials",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  }
}
```

Response headers

```
content-length: 169
content-type: application/json
date: Tue, 15 Sep 2020 06:38:50 GMT
server: uvicorn
```

DELETE:

DELETE

/admin/books/{book_id} Delete Book

Parameters

Cancel

Name	Description
book_id required	
Integer (path)	102

Execute

Response:

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8080/admin/books/102' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8080/admin/books/102

Server response

Code

Details

200

Response body

```
{
  "message": "Book deleted successfully",
  "book": {
    "id": 102,
    "title": "Database Management",
    "author": "Robert Brown",
    "price": 100,
    "isbn": "9789876543210",
    "status": "Available"
  }
}
```

Download

Response headers

```
content-length: 187
content-type: application/json
date: Tue, 25 Sep 2024 06:30:35 GMT
server: uvicorn
```